

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

UNIT AND MEASUREMENTS

LECTURE-5

PHYSICS

WAJAHID^{Sir}

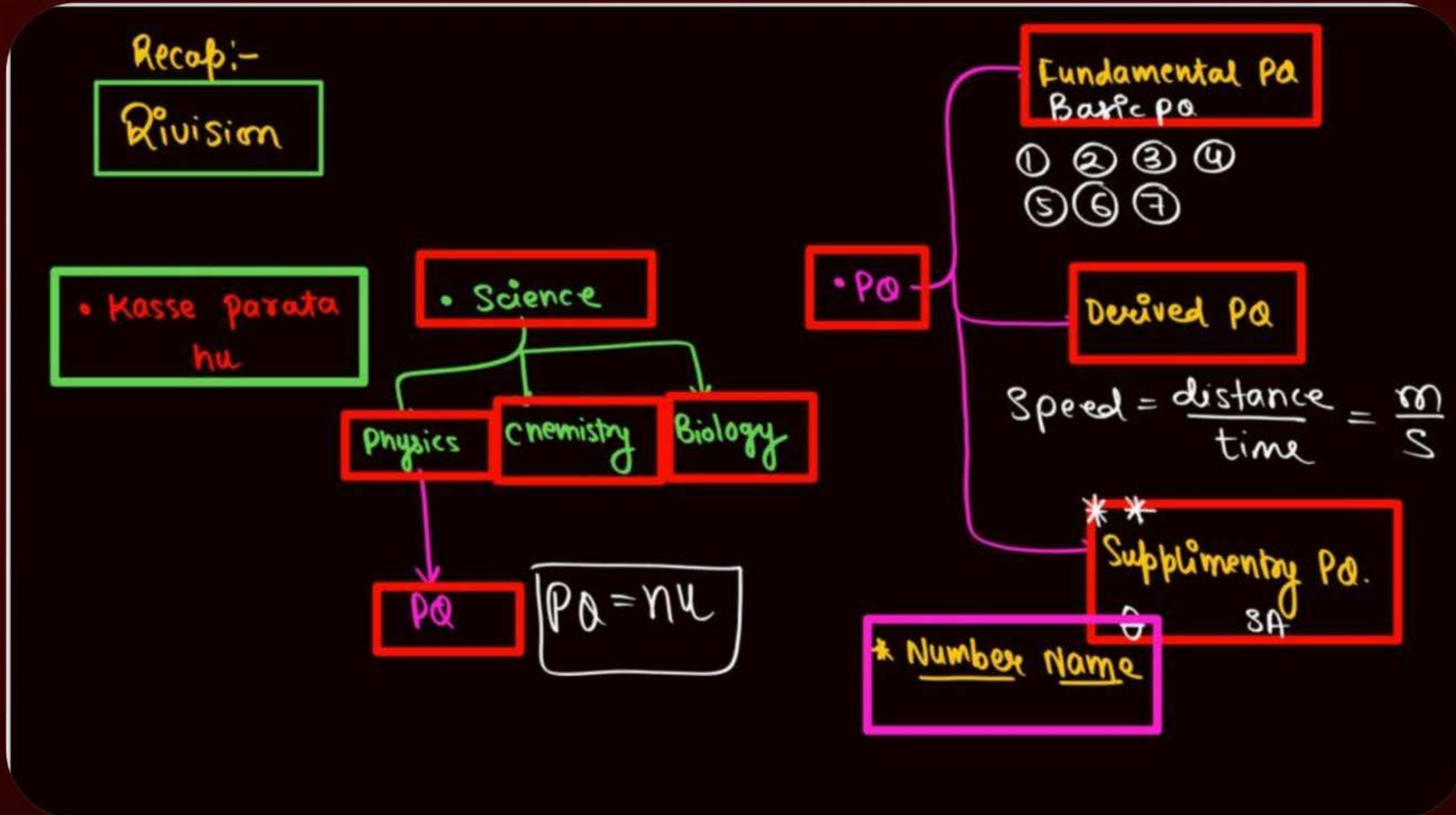


Topics To Covered

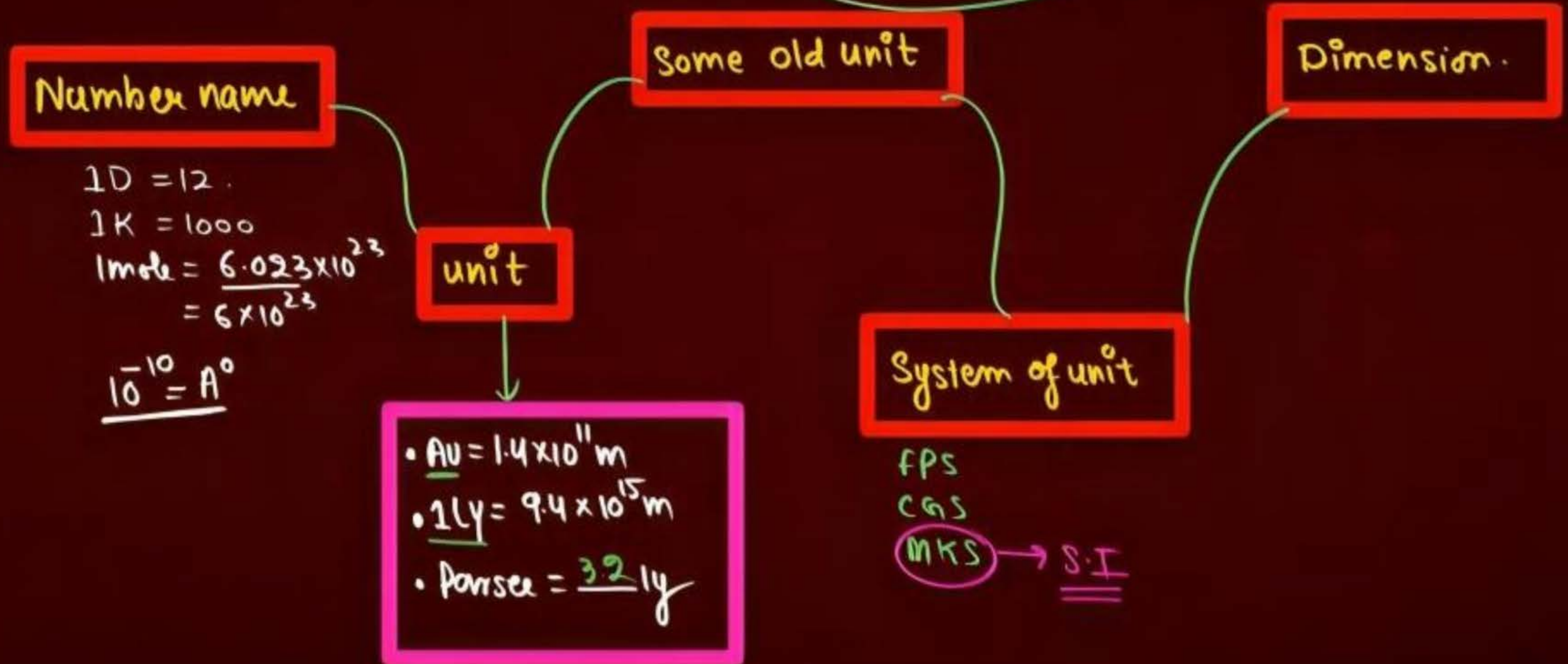
- 1** DIVISION
- 2** [C] TO DRIVE THE PHYSICS FORMULA
- 3** [D] TO CONVERT ONE SYSTEM OF UNIT INTO ANOTHER SYSTEM OF UNIT
- 4** LIMITATION OF DIMENSIONAL ANALYSIS

“The future belongs to
those who believe in
the beauty of their
dreams.”

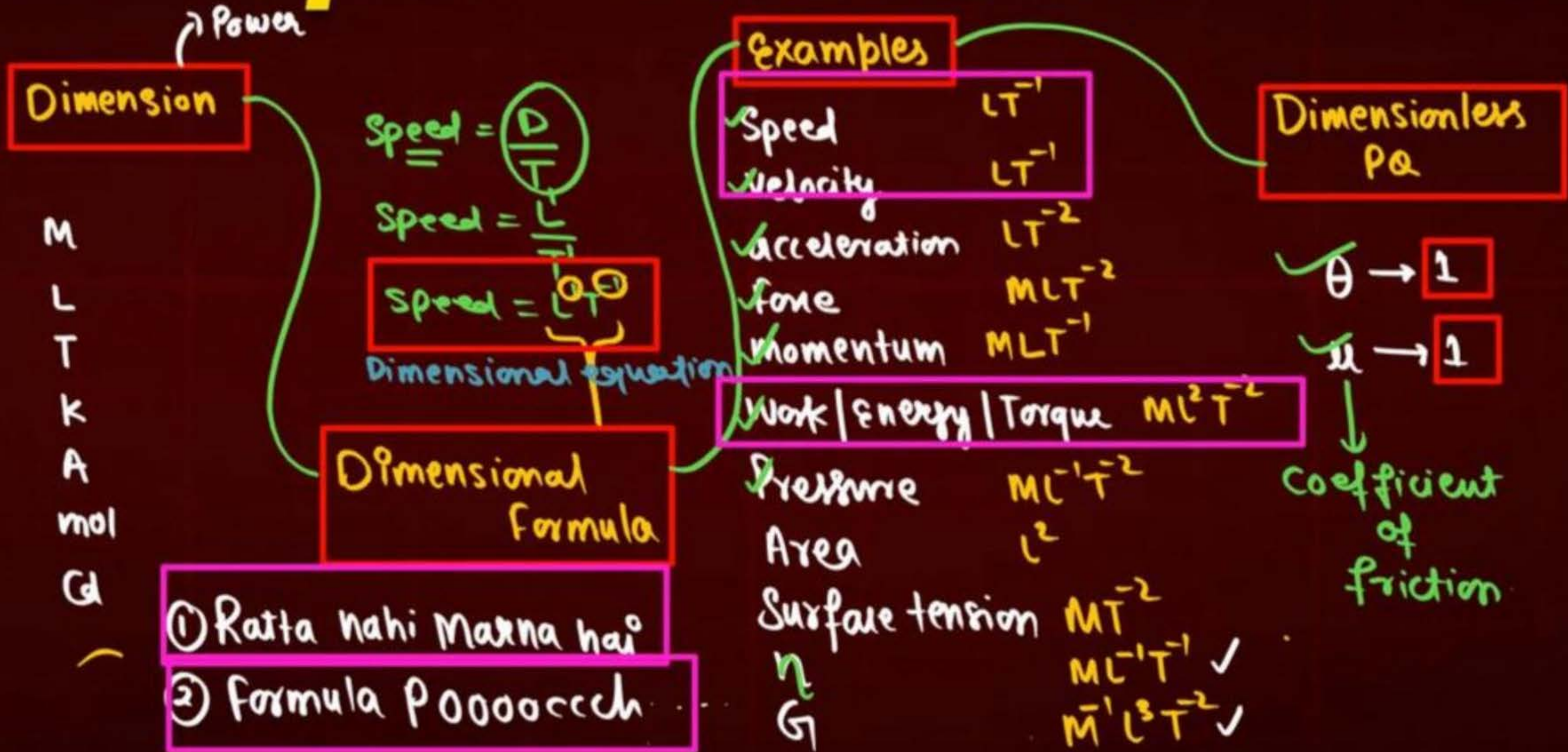
Recap From 1st Lec.



Recap From 2. Lec.



Recap From 3rd Lec.



Recap From 4th Lec.

Application
of
Dimension

a ✓

b ✓

c
d

① To find out the
Dimensional
Formula

$$\eta \rightarrow \text{ML}^{-1}\text{T}^{-1}$$

$$G \rightarrow \text{ML}^2\text{T}^{-2}$$

② To check the
Correctness of
Physics Formula

Law of
Homogeneity

$$x + x = 2x$$

$$2x - x = x$$

① $v = u + at$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\text{LT}^{-1} \quad \text{LT}^{-1} \quad \text{LT}^{-1}$$

② $s = ut + \frac{1}{2}at^2$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\text{L} \quad \text{LT}^{-1} \quad \text{LT}^{-1}$$

③ $v^2 = u^2 + 2as$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\text{L}^2\text{T}^{-2} \quad \text{L}^2\text{T}^{-2} \quad \text{LT}^{-2}$$

④ $F = \frac{mv^2}{r}$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$\text{MLT}^{-2} \quad \text{MLT}^{-2} \quad \text{MLT}^{-2}$$

QUESTIONS

H/w → class main samja raha hu



The velocity of a particle (v) at a instant (t) is given by $v = at + bt^2$ the dimension of b is:

2017

$a = ?$

$b = ?$

1 L

2 LT^{-1}

3 LT^{-2}

4 LT^{-3}

$\overset{D}{\underset{a}{\textcircled{v}}} = \overset{D}{\textcircled{at}} + \overset{D}{\textcircled{bt^2}}$

$$V = at$$

$$LT^{-1} = aT$$

$$\frac{LT^{-1}}{T} = a$$

$$LT^{-1}T^{-1} = a$$

$**$ $LT^{-2} = a$

$V = bt^2$ (velocity) $V = LT^{-1}$
(time) $t \rightarrow T$

$$LT^{-1} = bT^2$$

$$\frac{LT^{-1}}{T^2} = b$$

$$LT^{-1}T^{-2} = b$$

$$LT^{-1-2} = b$$

$$LT^{-3} = b$$

Q Dimensional formula of Volume = ?

Sol

$$\text{Volume} = L \times B \times h$$

$$V = L \times L \times L$$

$$V = (L^3)$$

$$\underline{A \rightarrow L^2}$$

Q. H/w

Find out the Dimensions of (a) and (b) in Vaander waal equation

✓ (2024) JKBOSE

P → Pressure

V → Volume

R → universal gas constt $V = L^3$

T → Temperature

$$\left(P + \frac{a}{V^2} \right) (V - b) = RT$$

Sol $a = ?$

$$P = \frac{a}{V^2}$$

$$ML^{-1}T^{-2} = \frac{a}{(L^3)^2}$$

$$ML^{-1}T^{-2} = \frac{a}{L^6}$$

$$ML^{-1}T^{-2}L^6 = a$$

$$ML^5T^{-2} = a$$

$$ML^{-1+6}T^{-2} = a$$

$$\boxed{ML^5T^{-2} = a}$$

$$P = \frac{F}{A} = \frac{MKT^{-2}}{L^2}$$

$$P = MT^{-2}L^{-1}$$

$$P = ML^{-1}T^{-2}$$

$$\left(p + \frac{a}{v^2}\right)(v - \underline{b}) = RT \quad 6^{\text{th}}(\underline{\text{math}})$$

$$(\underline{v} - \underline{b})$$

$$v = b$$
$$\boxed{v^3 = b} \quad \checkmark$$

$$v^2$$
$$\rightarrow (v^3)^2$$

$$v^3 \times v^3$$

$$v^{3+3}$$

$$v^6$$

HOMWORK

H/w Give the Dimensional formula of R (universal gas constt)

$$\underline{PV = nRT}$$

#

$$P = M L^{-1} T^{-2}$$

$$V \rightarrow L^3$$

$$\frac{PV}{nT} = R$$

$$\frac{M L^{-1} T^{-2} L^3}{\text{mol K}} = R$$

$$\frac{M L^2 T^{-2}}{\text{mol K}} = R$$

θ

$P \rightarrow$ Pressure

$V \rightarrow$ Volume

$n \rightarrow$ no. of mole

R

$T \rightarrow$ Temperature

$$R = \frac{M L^{-1} L^3 T^{-2}}{\text{mol K}}$$

$$R = \frac{M L^{-1+3} T^{-2}}{\text{mol K}}$$

$$R = M L^2 T^{-2} \text{mol}^{-1} \text{K}^{-1}$$

$$T = 2\pi \sqrt{l/g}$$

$$2T = 2 \times 2\pi \sqrt{l/g}$$

$$\cancel{2T} = \cancel{4\pi} \sqrt{l/g} \rightarrow \text{roota}$$

$g \rightarrow g/\text{am}$

$$T = 2\pi \sqrt{l/g}$$

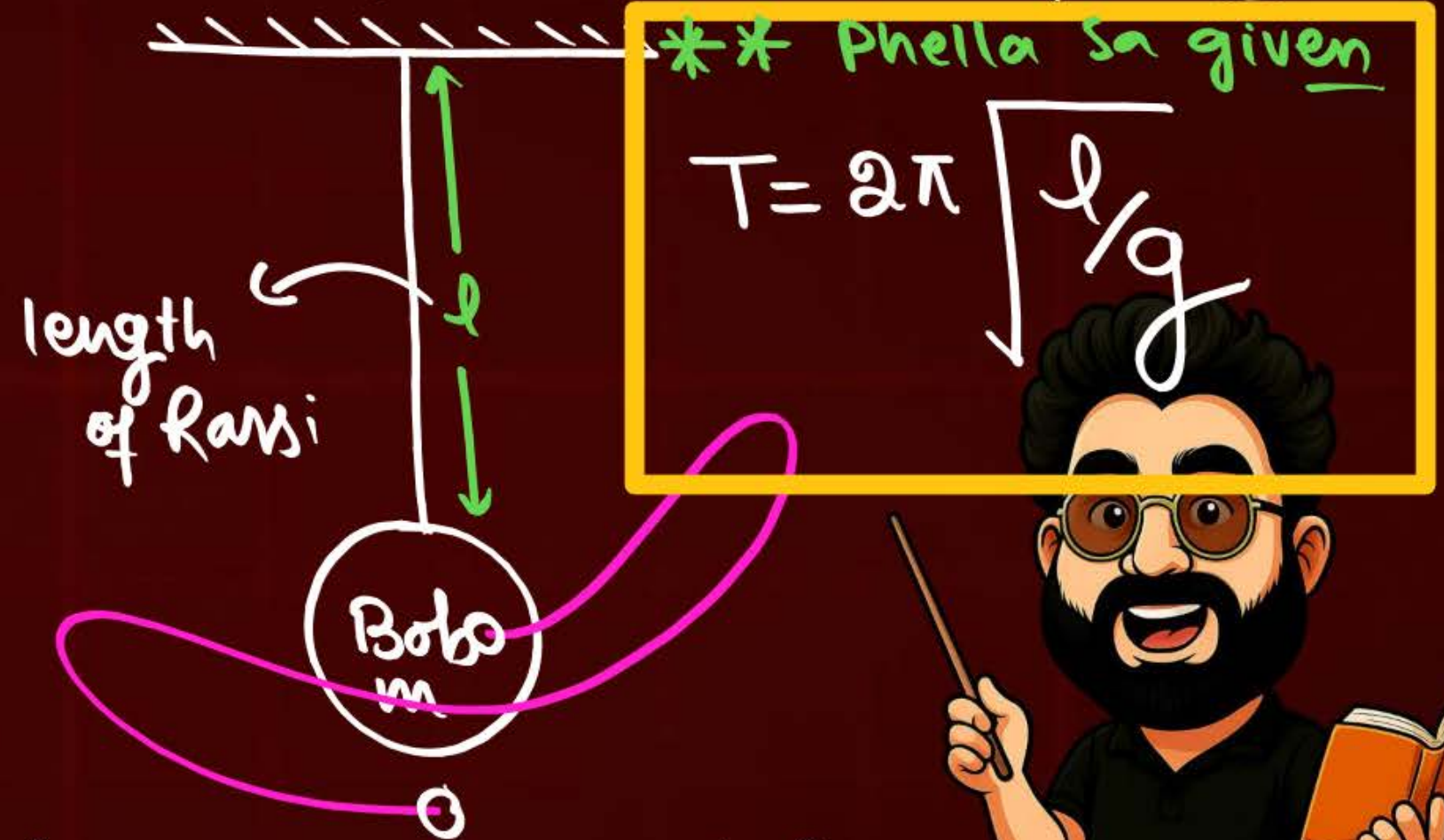
C) TO DRIVE THE PHYSICS FORMULA

Oscillation
Chapter (13)

- we also derive the physics formula Simple Pendulum using Dimension

Kasse = ? Tarika = ?

Time period of pendulum
using Dimensional Analysis



2 → number m → mass of bob

$\frac{22}{7} = 3.14 = \pi \rightarrow$ number T → Time period
g → all due to gravity



• Derive the Time period of pendulum using Dimensional Analysis

Let time period of pendulum depend on mass (m), length (l) and acc due to gravity (g)

$$T \rightarrow m^a l^b g^c \quad g \rightsquigarrow a = LT^{-2}$$

$$T \rightarrow m^a l^b g^c \quad \text{--- (1)}$$

$$M^0 L^0 T^1 \rightarrow M^a L^b (LT^{-2})^c$$

$$M^0 L^0 T^1 \rightarrow M^a L^b L^c T^{-2c}$$

$$\underbrace{M^0}_{\sim} \underbrace{L^0}_{\sim} \underbrace{T^1}_{\sim} \rightarrow \underbrace{M^a}_{\sim} \underbrace{L^{b+c}}_{\sim} \underbrace{T^{-2c}}_{\sim}$$

on comparing the powers
b/s

$$a = 0$$

$$b + c = 0$$

$$b + \frac{-1}{2} = 0$$

$$b = +\frac{1}{2}$$

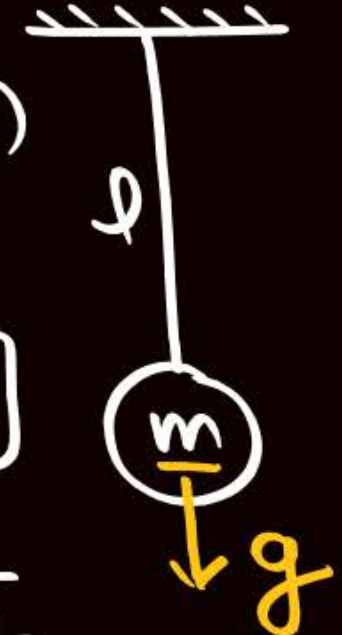
$$-2c = 1$$

$$c = \frac{1}{-2}$$

$$c = -\frac{1}{2}$$

Put value of a , b and c in equation (1)

$$T \rightarrow m^0 l^{1/2} g^{-1/2}$$



$$T = 2\pi \sqrt{l/g}$$

$$a^{1/2} = \sqrt{a}$$

$$T \rightarrow \underline{m^0} l^{1/2} g^{-1/2}$$

$$T \rightarrow 1 \times l^{1/2} g^{-1/2}$$

$$T \rightarrow l^{1/2} g^{-1/2}$$

$$T \rightarrow \frac{l^{1/2}}{g^{1/2}}$$

$$T \propto \frac{l^{1/2}}{g^{1/2}}$$

$$T \propto \frac{\sqrt{l}}{\sqrt{g}}$$

$$2^{1/2} = \sqrt{2}$$

$$l^{1/2} = \sqrt{l}$$

$$g^{1/2} = \sqrt{g}$$

$$T \propto \sqrt{l/g}$$

$$T = K \sqrt{l/g}$$

const

2π

$$T = 2\pi \sqrt{l/g}$$

Repeat $T \rightarrow m^a l^b g^c$ — ① $g = a = LT^{-2}$ $T \rightarrow m^0 l^{1/2} g^{-1/2}$

$$T \rightarrow M^a L^b (LT^{-2})^c$$

$$M^0 L^0 T^1 \rightarrow M^a L^b L^c T^{-2c}$$

$$M^0 L^0 T^1 \rightarrow M^a L^{b+c} T^{-2c}$$

$c \rightarrow \text{power}$

$$\underline{a=0}$$

$$a=0$$

$$b+c=0$$

$$b + -\frac{1}{2} = 0$$

$$b = \frac{1}{2}$$

$$-2c = 1$$

$$c = \frac{1}{-2}$$

$$c = -\frac{1}{2}$$

Put in equation 1st

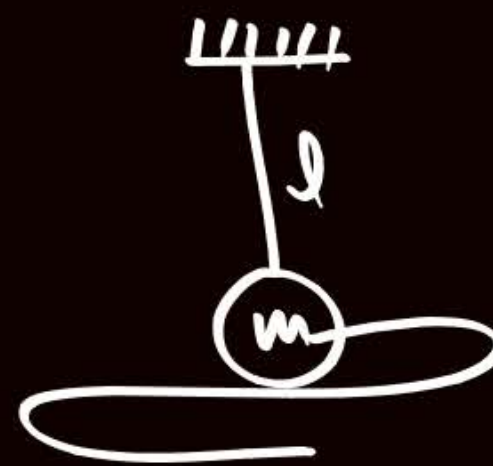
$$T \rightarrow m^0 l^{1/2} g^{-1/2}$$

$$T \rightarrow 1 l^{1/2} g^{-1/2}$$

$$T \rightarrow \frac{l^{1/2}}{g^{1/2}}$$

$$T \propto \frac{l^{1/2}}{g^{1/2}}$$

$$T \propto \frac{\sqrt{l}}{\sqrt{g}}$$



$$T = 2\pi \sqrt{l/g}$$

$$T = 2\pi \frac{l^{1/2}}{g^{1/2}}$$

$$T \propto \sqrt{l/g}$$

$$T = K \sqrt{l/g}$$

$$T = 2\pi \sqrt{l/g}$$

$$K.E = \frac{1}{2} m^{(1)} v^{(2)}$$

Q DRIVE THE STROKE LAW USING DIMENSIONAL ANALYSIS



$$\bullet \textcircled{F} = 6\pi\eta r v$$

$F \rightarrow \text{Force (MLT}^{-2}\text{)}$
 $6 \rightarrow \text{no, } \pi = 3.14$

η (coefficient of viscosity) = $\text{ML}^{-1}\text{T}^{-1}$

$r \rightarrow \text{radius (L)}$

$v \rightarrow \text{velocity (LT}^{-1}\text{)}$

$$\textcircled{F} \rightarrow \eta^a r^b v^c \quad \text{--- ①}$$

$$\underline{\text{MLT}^{-2}} \rightarrow (\underline{\text{ML}^{-1}\text{T}^{-1}})^a (\underline{\text{L}})^b (\underline{\text{LT}^{-1}})^c$$

$$\underline{\text{MLT}^{-2}} \rightarrow \underline{\text{M}^a} \underline{\text{L}^{-a}} \textcircled{\underline{\text{T}^{-a}}} \underline{\text{L}^b} \underline{\text{L}^c} \textcircled{\underline{\text{T}^{-c}}}$$

$$\underline{\text{MLT}^{-2}} \rightarrow \underline{\text{M}^a} \underline{\text{L}^{-a} \text{L}^b \text{L}^c} \underline{\text{T}^{-a} \text{T}^{-c}}$$

$$\underline{\text{MLT}^{-2}} \rightarrow \underline{\text{M}^a} \underline{\text{L}^{-a+b+c}} \underline{\text{T}^{-a-c}}$$



$$\underline{M}^{\textcircled{1}} \underline{L}^{\textcircled{1}} \underline{T}^{\textcircled{-2}} \rightarrow \underline{M}^{\textcircled{a}} \underline{L}^{\textcircled{-a+b+c}} \underline{T}^{\textcircled{-a-c}}$$

compare the power

$$\textcircled{a=1}$$

$$-a+b+c=1$$

$$-1+b+1=1$$

$$b+1-1=1$$

$$b+0=1$$

$$\boxed{b=1}$$

$$-a-c=-2$$

$$-1-c=-2$$

$$-1+2=c$$

$$+2-1=c$$

$$\boxed{1=c}$$

$$c=1$$

Put the value of
 $a=1, b=1, c=1$

$$F \rightarrow \eta^a \tau^b \nu^c$$

$$F \rightarrow \eta' \tau' \nu'$$

$$f \propto \eta \tau \nu$$

$$f = \textcircled{k}^{\textcircled{6\pi}} \eta \tau \nu$$

$$\boxed{f = 6\pi \eta \tau \nu}$$

u/w

$$T = 2\pi \sqrt{l/g}$$

$$F = 6\pi\eta r v$$

SUMMARY



3Q

Revision
 $L-1 \rightsquigarrow L-4$

© To derive the PF

$$T = 2\pi \sqrt{l/g}$$

$$F_v = 6\pi\eta r v$$

$$H/\omega$$

HOMework



Q. Find the Dimensional formula of charge Q

Sol

Class

10th

$$\underline{I} = \frac{Q}{t}$$

Thank
You